Plotting Data with Pandas

Introducing the Plotting Ecosystem

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Overview

Plotting tools and packages in Python

The matplotlib package

Establishing the plotting environment

What Will **Not** be Covered

This course is not an introduction to Python programming

This course does not focus on the art or science of data visualization

The focus of this course is <u>how</u> to plot data

Prerequisites

Basic knowledge of Python and pandas

Understanding of data frames

Plotting Ecosystem in Python

matplotlib

All plots created in pandas are matplotlib objects

seaborn

Seaborn is a package focused on statistical visualization

bokeh

Bokeh is used to created interactive visualizations for the Internet

plotly

Plotly can create interactive and embeddable visualizations

Introducing matplotlib

Pyplot is a matplotlib submodule that provides functions for working with different plot elements

The syntax of matplotlib and pyplot is closely related to the MATLAB statistical software

Jupyter Notebook

Jupyter Notebook is an interactive web application to easily create and run code

http://www.jupyter.org

Check out the Pluralsight course:
Getting Started with Jupyter Notebook, Douglas Starnes

Anaconda

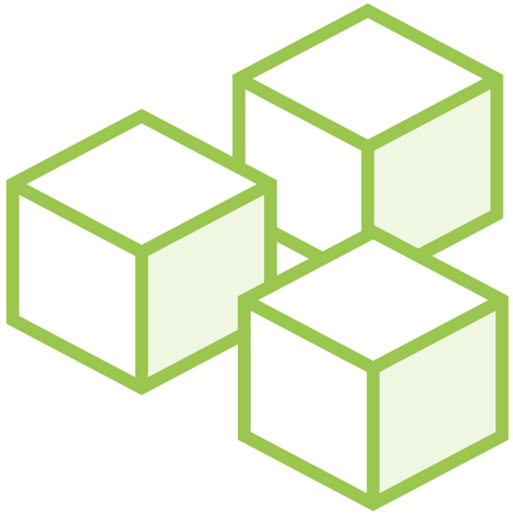
Python distribution

Preloaded with pandas, matplotlib, Jupyter Notebook, and other tools

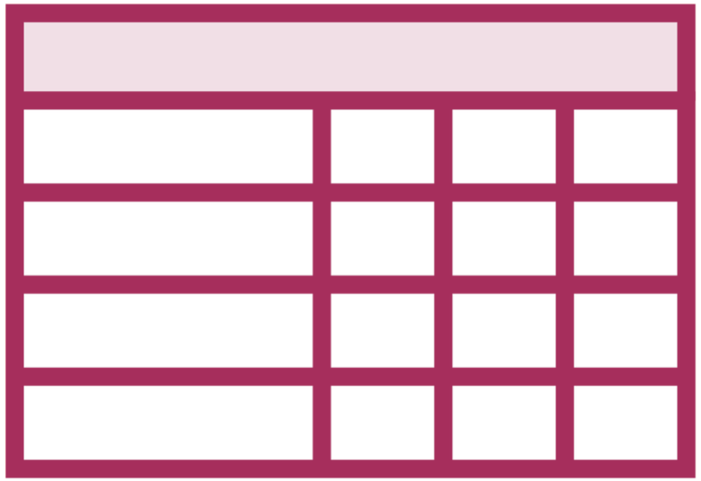
Free download at www.anaconda.com

ANACONDA.

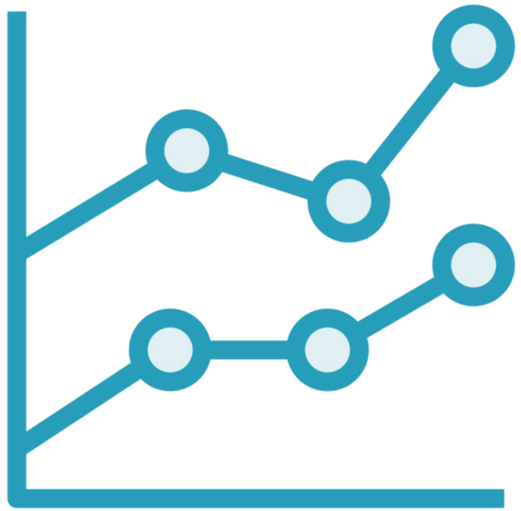
Creating a Plot



Import the necessary packages

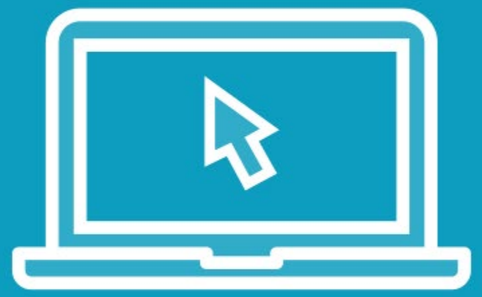


Import or create data



Call the appropriate plotting functions

Demo



Establishing our plotting environment

Summary

Matplotlib is the fundamental data visualization library

Three-step framework for creating a plot

- Importing the necessary packages
- Preparing our data
- Running the appropriate plotting functions